Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Substitution: power = $P = \frac{2.50}{10}$ (1) Power = 0.25 [kW] (1)	1	1		2	2	
		(ii)	I	75 [p]		1		1		
			II	Substitution: $\frac{75\ \textbf{ecf}}{2.50}$ (1) Cost of 1 unit = 30 [p]	1	1		2	2	
	(b)			To reduce use of fossil fuels / to reduce {greenhouse gas / carbon / CO_2} emissions / to prevent power cuts / to reduce {carbon footprint / climate change / global warming / greenhouse effect} Accept reduce pollution / to help the environment		1		1		
	(c)			**Indicative content:** The earth wire protects users of electricity. The earth wire provides a low resistance path to earth if a fault causes a metal cased appliance to become live. A fuse is used to protect circuits from overheating. If there is a short circuit causing too much current, the fuse melts. The rccb circuit breakers protect users of electricity. The rccb switches off the circuit if there is a difference between the currents in the live and neutral wires.	6			6		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				5–6 marks Correctly describes the fault that causes all 3 safety features to work and what they are used to protect. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3–4 marks** Correctly describes the fault that causes 2 safety features to work and what they are used to protect **OR** limited attempt at all 3. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1–2 marks** Correctly describes the fault that causes 1 safety feature to work and what it is used to protect **OR** limited attempt at 1 or 2. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure*. *The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*						
				Question 5 total	**8**	**4**	**0**	**12**	**4**	**0**